National College of Computer Studies

Paknajol, Kathmandu

Lab Report

On

Information Security

Submitted by:

Name: Siddhartha Shakya

Program: BIM

Roll no: 18

Section: B

Submitted to:

Radha Krishna Gajurel

National College of Computer Studies

Paknajol, Kathmandu

Lab Report

On

Artificial Intelligence

Submitted by:

Name: Siddhartha Shakya

Program: BIM

Roll no: 18

Section: B

Submitted to:

Er.Khushbu Kumari Sarraf

S.N.	Program Title	Submission Date	Signature	Remarks

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 AES.py
Encrypted: H3rCvTSbhlBq1cgOmg9VOAdnaa89UralQHeDX0swecbG4WEsZtSCFc9MtuYrmtbK
Decrypted: My name is Siddhartha
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 congruence_modulo.py
Enter first number (a): 12
Enter second number (b): 5
Enter modulus (n): 7
12 ≡ 5 (mod 7)
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 RSA.py
Enter message [less than 187]: 10
Public Key: (7, 187)
Private Key: (23, 187)
Encrypted: 175
Decrypted: 10
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 diffie_hellman.py
Enter a prime number (p): 37
Enter a primitive root (g): 5
Enter Alice's private key: 3
Enter Bob's private key: 6

Public Values:
Alice sends: 14
Bob sends: 11

Shared Secret (Alice): 36
Shared Secret (Bob): 36

Key Exchange Successful!
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 Greedy.py
Traversal:
A C G
Final Path: A → C → G
Total h(n) along path: 5
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 astar.py
Traversal:
A C B E F
Final Path: A → B → E → F
Total Cost: 4
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 DLS.py
Traversal:
A B D H E I

Goal found within depth limit
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 DLS.py
Traversal:
A B D E C F G

Goal not found within depth limit
```

```
pritishmashakya@Pritishmas-MacBook-Pro Siddhartha_Lab2 % python3 IDDFS.py

Depth Limit = 0
A
Depth Limit = 1
A B C
Depth Limit = 2
A B D E C F G
Depth Limit = 3
A B D H E I
```

S.N.	Title	Submission Date	Signature	Remarks

National College of Computer Studies

Paknajol, Kathmandu

Lab Report No. 3

Programming with Python

Submitted by:

Name: Siddhartha Shakya

Program: BIM

Roll no: 18

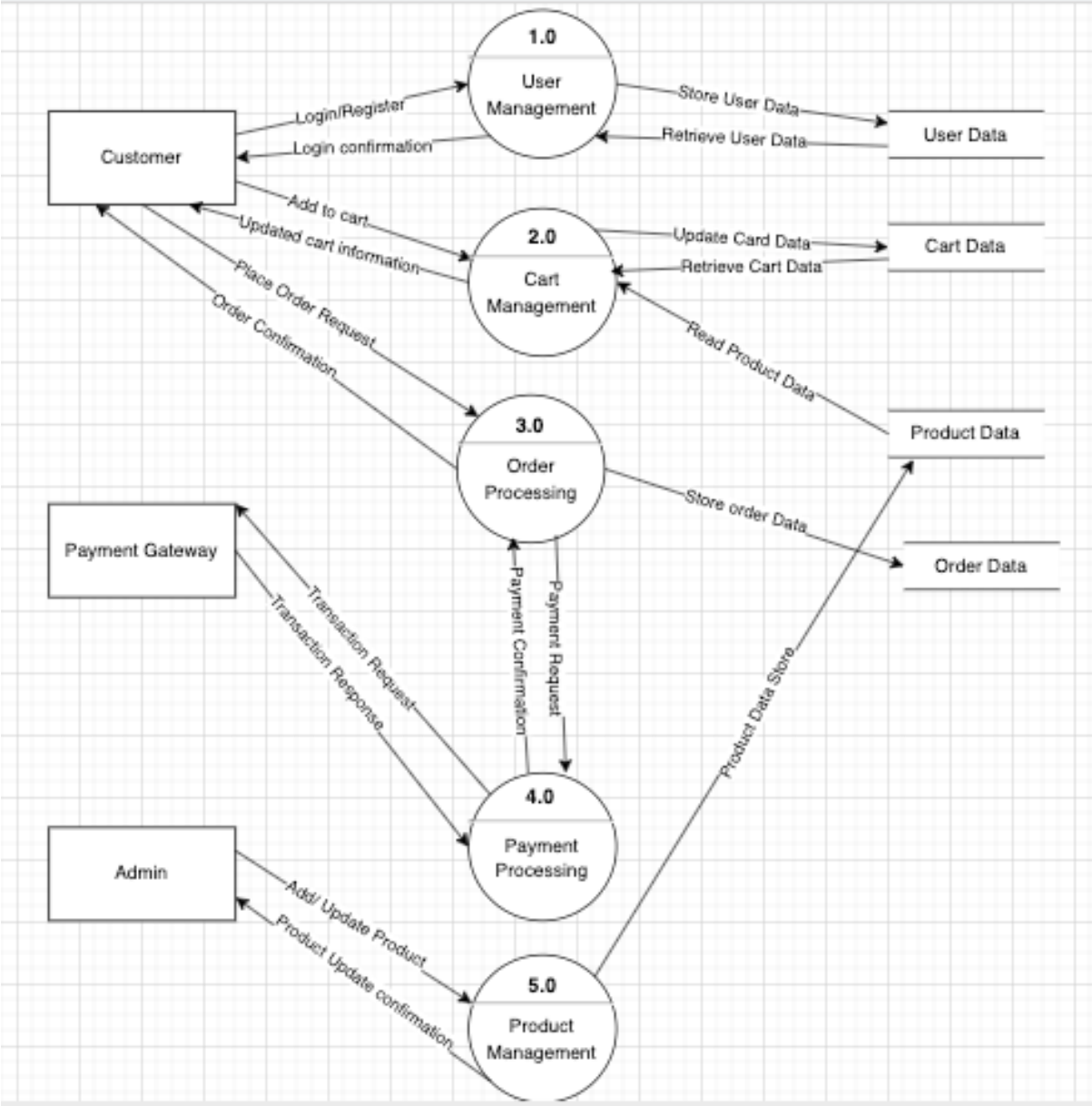
Section: B

Submitted to:

Sumit Ghising

Date of submission: 2026/02/05

Level 0 DFD for an Online Shopping System



Level 1 DFD for an online Shopping System